Unit 1 – Using Objects and Methods Name: _____
HW 4

1) Complete the following table by predicting the output of each statement.

	Code	Expected Output
a)	`System.out.println(7 / 2);`	
b)	`System.out.println(7.0 / 2);`	
c)	`System.out.println((double) 7 / 2);`	
d)	`System.out.println((double) (7 / 2));`	

2) Explain, in your own words, why there is a difference between `(double) 7 / 2` and `(double) (7 / 2)`.

3) The following code contains an error.

```
int x = (double) 7 / 2;
```

a) What type of error is this?

b) Why is this an issue?

c) How can you fix this issue?

4) Complete the following table by predicting the output of each statement.

	Code	Expected Output

a)	`double value = 5.7;`
	`int x = (int) (value + 0.5);`
	`System.out.println(x);`	

b)	`double value = 2.3;`
	`int y = (int) (value + 0.5);`
	`System.out.println(y);`	

c)	`double value = 8.9;`
	`int z = (int) (value + 0.5);`
	`System.out.println(z);`	

5) Imagine you are writing a grading program that needs to round students' average scores.

a) Why would `int roundedScore = (int) (score + 0.5);` be a useful trick?

b) What would happen if we just used `(int) score;` instead?

6) Consider the code segment below.

```
int num = 5/2;
```

Circle or highlight the statements that are equivalent to the code segment above.

I. `int num = (int)(5/2);`

II. `int num = (int)(5.0/2);`

III. `int num = 5.0/2.0;`

IV. `int num = (int) 5/2;`

V. `int num = (double) 5 / (double) 2;`

7) Determine the final value of the expressions below.

a) `8 / 3 + (double) 5 / 2`

b) `(double) 10 / 4 + 3 / 2`

c) `10.5/2 + (int) 4.9 + 3.25`

d) `20.0/4 + (double) (15/2) - (int) 5.5`